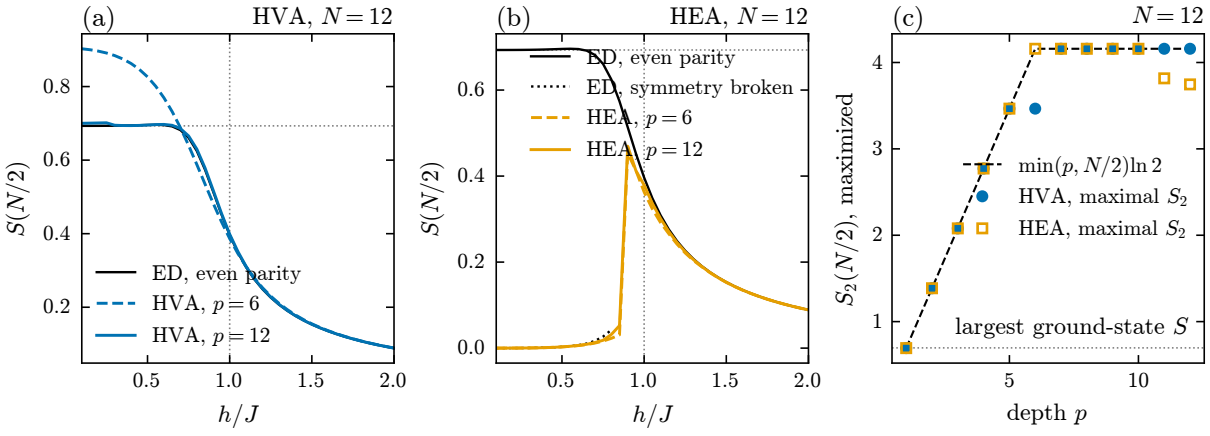
(a)
HVA, $N = 12$
$S(N/2)$
0.8
0.6
0.4
0.2
ED, even parity
HVA, $p = 6$
HVA, $p = 12$
0.5
1.0
1.5
2.0
h/J
(b)
HEA, $N = 12$
$S(N/2)$
0.6
0.4
0.2
0.0
ED, even parity
ED, symmetry broken
HEA, $p = 6$
HEA, $p = 12$
0.5
1.0
1.5
2.0
h/J
(c)
$N = 12$
$S_2(N/2)$, maximized
4
3
2
1
$\min(p, N/2)\ln 2$
HVA, maximal S_2
HEA, maximal S_2
largest ground-state S
5
10
depth p